MAE5005 Computational Fluid Dynamics, Spring 2026

Homework Set 7

Due Monday, May 11, 2026 by 19:00, by online submission through *Blackboard*

19. (30 points) Revisit Problems 13, 17 and 18.

    (a) For the same initial condition as stated in Problem 13, $C \equiv a\Delta t/\Delta x = 1.1$ and $N = 160$, study how the L1 and L2 errors vary with time for the following five schemes: (a) the upwind scheme, (b) the downwind scheme, (c) the central scheme; (d) the Lax-Wendroff scheme, and (e) Scheme B.

    (b) For each scheme which becomes eventually unstable, compute the corresponding amplitude growth rate using the L1 and L2 data from your code.

    (c) Can you predict, theoretically, each amplitude growth rate that you observed in Part (b)?

    Note: This problem has a lot of similarity to one of the midterm exam problems.

20. (40 points) Now consider the three schemes: the upwind scheme, the Lax-Wendroff scheme, and Scheme B, and the same initial condition. Write your codes for these schemes in double precision representation. Our purpose here is to compare and understand how the truncation errors vary with the grid spacing Δx and C.

    (a) According to the modified equations for the three schemes, state how the leading-order truncation error for each scheme would vary with Δx and C.

    (b) For a fixed $C = 0.1$, plot the actual $L1$ and $L2$ errors for the three schemes as a function of Δx, for $N = 20, 40, 80, 160, 320$, at several representative times.

    (c) For a fixed $N = 160$, plot the actual $L1$ and $L2$ errors for the three schemes as a function of C, for $0 < C \leq 1.0$, at several representative times.

    (d) Do you best, to theoretically predict the observed $L1$ and $L2$ errors in Parts (b) and (c).

    Note: We will discuss the results for these problems when we meet again on May 11.